MATRIX ADDITION AND MULTIPLICATION

LINEAR ALGEBRA

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Matrix

Adding Matrices

Multiplying Matrices

- Scalar and Vector

MATRIX

A **rectangular array** of numbers, symbols, or expressions, arranged in **rows** and **columns**

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

DIMENSION

The dimension of a matrix is the number of **rows** and **columns**

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

$$m \times n$$

DIMENSION

The dimension of a matrix is the number of **rows** and **columns**

$$\boldsymbol{m \times n}$$

2 × 3 Matrix

$$A = \begin{bmatrix} 5 & -3 & 0 \\ 1 & 9 & -4 \end{bmatrix}$$

3 × 2 Matrix

$$B = \begin{bmatrix} 5 & 2 \\ 1 & 0 \\ 8 & 8 \end{bmatrix}$$

The elements of a matrix are the **entries**

a_{ij}

2×3 Matrix

$$A = \begin{bmatrix} \overset{a_{11}}{5} & \overset{a_{12}}{-3} & \overset{a_{13}}{0} \\ \overset{a_{21}}{1} & \overset{a_{22}}{9} & \overset{a_{23}}{-4} \end{bmatrix}$$

3×2 Matrix

$$B = \begin{bmatrix} \overset{b_{11}}{5} & \overset{b_{12}}{2} \\ \overset{b_{21}}{1} & \overset{b_{22}}{0} \\ \overset{b_{31}}{8} & \overset{b_{32}}{8} \end{bmatrix}$$

ADDING MATRICES

Same Dimensions

Matrices must have the same number of rows and columns

$$A = \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

Element-wise Addition

Adding elements in the **same position** from both matrices

$$B = \begin{bmatrix} 3 & 5 & -7 \\ 1 & 2 & 1 \end{bmatrix}$$

Determine $\boldsymbol{C} = \boldsymbol{A} + \boldsymbol{B}$

$$\boldsymbol{C} = \begin{bmatrix} & & \\ & & \end{bmatrix}$$

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

$$\boldsymbol{B} = \begin{bmatrix} 3 & 5 & -7 \\ 1 & 2 & 1 \end{bmatrix}$$

Determine $\boldsymbol{C} = \boldsymbol{A} + \boldsymbol{B}$

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

$$\boldsymbol{B} = \begin{bmatrix} 3 & 5 & -7 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\boldsymbol{C} = \begin{bmatrix} 4 & & \\ & & \\ & & \end{bmatrix}$$

Element-wise addition,

$$\boldsymbol{C_{11}} = \boldsymbol{a_{11}} + \boldsymbol{b_{11}}$$

$$\boldsymbol{C_{11}} = \boldsymbol{1} + \boldsymbol{3}$$

$$\boldsymbol{C_{11}} = \boldsymbol{4}$$

Determine $C = A + B$

$$A = \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 5 & -7 \\ 1 & 2 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 4 & 5 & \\ & & \\ & & \end{bmatrix}$$

Element-wise addition,

$$C_{12} = a_{12} + b_{12}$$

$$C_{12} = 0 + 5$$

$$C_{12} = 5$$

Determine $C = A + B$

$$A = \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 5 & -7 \\ 1 & 2 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 4 & 5 & -11 \\ & & \\ & & \end{bmatrix}$$

Element-wise addition,

$$C_{13} = a_{13} + b_{13}$$

$$C_{13} = -4 + (-7)$$

$$C_{13} = -11$$

Determine $C = A + B$

$$A = \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 5 & -7 \\ 1 & 2 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 4 & 5 & -11 \\ 6 & & \end{bmatrix}$$

Element-wise addition,

$$C_{21} = a_{21} + b_{21}$$

$$C_{21} = 5 + 1$$

$$C_{21} = 6$$

Determine $C = A + B$

$$A = \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 5 & -7 \\ 1 & 2 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 4 & 5 & -11 \\ 6 & 11 & \end{bmatrix}$$

Element-wise addition,

$$C_{22} = a_{22} + b_{22}$$

$$C_{22} = 9 + 2$$

$$C_{22} = 11$$

Determine $\boldsymbol{C} = \boldsymbol{A} + \boldsymbol{B}$

$$\boldsymbol{A} = \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

$$\boldsymbol{B} = \begin{bmatrix} 3 & 5 & -7 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\boldsymbol{C} = \begin{bmatrix} 4 & 5 & -11 \\ 6 & 11 & 2 \end{bmatrix}$$

Element-wise addition,

$$\boldsymbol{C_{23}} = \boldsymbol{a_{23}} + \boldsymbol{b_{23}}$$

$$\boldsymbol{C_{23}} = \boldsymbol{1} + \boldsymbol{1}$$

$$\boldsymbol{C_{23}} = \boldsymbol{2}$$

Determine $C = A - B$

$$A = \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 5 & -7 \\ 1 & 2 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} -2 & -5 & 3 \\ 4 & 7 & 0 \end{bmatrix}$$

Matrices with **different dimensions** cannot be added or subtracted

PROPERTIES OF MATRIX ADDITION

Addition is associative

$$(\boldsymbol{A}+\boldsymbol{B})+\boldsymbol{C}=\boldsymbol{A}+(\boldsymbol{B}+\boldsymbol{C})$$

Additive identity (the zero matrix)

$$\boldsymbol{A}+[\mathbf{0}]=\boldsymbol{A}$$

Addition is commutative

$$\boldsymbol{A}+\boldsymbol{B}=\boldsymbol{B}+\boldsymbol{A}$$

MULTIPLYING MATRICES

SCALAR MULTIPLICATION

$$3 \cdot \begin{bmatrix} 1 & 0 & -4 \\ 5 & 9 & 1 \end{bmatrix}$$

Scalar is a **single** numerical value

$$\begin{bmatrix} 3 & 0 & -12 \\ 15 & 27 & 3 \end{bmatrix}$$

PROPERTIES OF SCALAR MULTIPLICATION

Scalar multiplication is distributive

$$r(A + B) = rA + rB$$

Scalar multiplication is associative

$$(r \cdot l)A = r(l \cdot A)$$

Multiplication by 0

$$0 \cdot A = [0]$$

Scalar multiplication is commutative

$$r \cdot A = A \cdot r$$

Row Vector

A matrix that has only **one row**

$1 \times n$ matrix

$$\vec{w} = \begin{bmatrix} w_1 & w_2 & w_3 & \cdots & w_n \end{bmatrix}$$

Column Vector

A matrix that has only **one column**

$n \times 1$ matrix

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_n \end{bmatrix}$$

ROW AND COLUMN VECTOR

Row Vector

A matrix that has only **one row**

$\mathbf{1 \times 4}$ matrix

$$\vec{w} = \begin{bmatrix} 3 & 6 & 1 & -1 \end{bmatrix}$$

Column Vector

A matrix that has only **one column**

$\mathbf{4 \times 1}$ matrix

$$\vec{v} = \begin{bmatrix} 1 \\ -2 \\ 3 \\ 5 \end{bmatrix}$$

VECTOR MULTIPLICATION

Multiplying a $1 \times n$ **row vector** with an $n \times 1$ **column vector**

$$\sum_{i=1}^{n} \vec{w}_i \vec{v}_i = v_1 w_1 + v_2 w_2 + \cdots w_n v_n$$

Determine $\vec{w} \cdot \vec{v}$

$$\sum_{i=1}^{4} \vec{w}_i \vec{v}_i = \vec{w}_1\vec{v}_1 + \vec{w}_2\vec{v}_2 + \vec{w}_3\vec{v}_3 + \vec{w}_4\vec{v}_4$$

$$\vec{w} = \begin{bmatrix} 3 & 6 & 1 & -1 \end{bmatrix}$$

$$= (3 \cdot 1) + (6 \cdot -2) + (1 \cdot 3) + (-1 \cdot 5)$$

$$= 3 - 12 + 3 - 5$$

$$= -11$$

$$\vec{v} = \begin{bmatrix} 1 \\ -2 \\ 3 \\ 5 \end{bmatrix}$$

$$\underset{m \times r}{\overset{A}{\begin{bmatrix} \\ i \\ \\ \end{bmatrix}}} \cdot \underset{r \times n}{\overset{B}{\begin{bmatrix} & j & \\ & & \\ & & \end{bmatrix}}} = \underset{m \times n}{\overset{C}{\begin{bmatrix} & & \\ & c_{ij} & \\ & & \end{bmatrix}}}$$

columns of A = **rows** of B

dot product

of ith row of A and j^{th} column of B

Determine $\boldsymbol{C} = \boldsymbol{A} \cdot \boldsymbol{B}$

$$\underset{4 \times 2}{\overset{\boldsymbol{A}}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \underset{2 \times 3}{\overset{\boldsymbol{B}}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \underset{4 \times 3}{\overset{\boldsymbol{C}}{\begin{bmatrix} \quad & \quad & \quad \\ \quad & \quad & \quad \\ \quad & \quad & \quad \\ \quad & \quad & \quad \end{bmatrix}}}$$

Determine $C = A \cdot B$

$$\overset{A}{\underset{4\times 2}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \overset{B}{\underset{2\times 3}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \overset{C}{\underset{4\times 3}{\begin{bmatrix} 11 & & \\ & & \\ & & \\ & & \end{bmatrix}}}$$

Dot Product,

$$C_{11} = \overrightarrow{a_1} \cdot \overrightarrow{b_1}$$

$$C_{11} = (2 \cdot 4) + (3 \cdot 1)$$

$$C_{11} = 11$$

Determine $C = A \cdot B$

$$\overset{A}{\underset{4\times 2}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \overset{B}{\underset{2\times 3}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \overset{C}{\underset{4\times 3}{\begin{bmatrix} 11 & 0 & \\ & & \\ & & \\ & & \end{bmatrix}}}$$

Dot Product,

$$C_{12} = \overrightarrow{a_1} \cdot \overrightarrow{b_2}$$

$$C_{12} = (2 \cdot 3) + (3 \cdot -2)$$

$$C_{12} = 0$$

Determine $C = A \cdot B$

$$\underset{4 \times 2}{\overset{A}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \underset{2 \times 3}{\overset{B}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \underset{4 \times 3}{\overset{C}{\begin{bmatrix} 11 & 0 & 21 \\ & & \\ & & \\ & & \end{bmatrix}}}$$

Dot Product,

$$C_{13} = \overrightarrow{a_1} \cdot \overrightarrow{b_3}$$

$$C_{13} = (2 \cdot 6) + (3 \cdot 3)$$

$$C_{13} = 21$$

Determine $C = A \cdot B$

$$\overset{A}{\underset{4\times 2}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \overset{B}{\underset{2\times 3}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \overset{C}{\underset{4\times 3}{\begin{bmatrix} 11 & 0 & 21 \\ -1 & & \\ & & \\ & & \end{bmatrix}}}$$

Dot Product,

$$C_{21} = \overrightarrow{a_2} \cdot \overrightarrow{b_1}$$

$$C_{21} = (1 \cdot 4) + (-5 \cdot 1)$$

$$C_{21} = -1$$

Determine $C = A \cdot B$

$$\overset{A}{\underset{4\times 2}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \overset{B}{\underset{2\times 3}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \overset{C}{\underset{4\times 3}{\begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & \\ & & \\ & & \end{bmatrix}}}$$

Dot Product,

$$C_{22} = \overrightarrow{a_2} \cdot \overrightarrow{b_2}$$

$$C_{22} = (1 \cdot 3) + (-5 \cdot -2)$$

$$C_{22} = 13$$

Determine $C = A \cdot B$

$$\underset{4 \times 2}{\overset{A}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \underset{2 \times 3}{\overset{B}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \underset{4 \times 3}{\overset{C}{\begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \\ & & \\ & & \end{bmatrix}}}$$

Dot Product,

$$C_{23} = \overrightarrow{a_2} \cdot \overrightarrow{b_3}$$

$$C_{23} = (1 \cdot 6) + (-5 \cdot 3)$$

$$C_{23} = -9$$

Determine $C = A \cdot B$

$$\underset{4 \times 2}{\overset{A}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \underset{2 \times 3}{\overset{B}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \underset{4 \times 3}{\overset{C}{\begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \\ 7 & & \\ & & \end{bmatrix}}}$$

Dot Product,

$$C_{31} = \overrightarrow{a_3} \cdot \overrightarrow{b_1}$$

$$C_{31} = (0 \cdot 4) + (7 \cdot 1)$$

$$C_{31} = 7$$

Determine $C = A \cdot B$

$$\overset{A}{\underset{4\times 2}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \overset{B}{\underset{2\times 3}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \overset{C}{\underset{4\times 3}{\begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \\ 7 & -14 & \\ & & \end{bmatrix}}}$$

Dot Product,

$$C_{32} = \overrightarrow{a_3} \cdot \overrightarrow{b_2}$$

$$C_{32} = (0 \cdot 3) + (7 \cdot -2)$$

$$C_{32} = -14$$

Determine $C = A \cdot B$

$$\overset{A}{\underset{4\times 2}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \overset{B}{\underset{2\times 3}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \overset{C}{\underset{4\times 3}{\begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \\ 7 & -14 & 21 \end{bmatrix}}}$$

Dot Product,

$$C_{33} = \overrightarrow{a_3} \cdot \overrightarrow{b_3}$$

$$C_{33} = (0 \cdot 6) + (7 \cdot 3)$$

$$C_{33} = 21$$

Determine $C = A \cdot B$

$$\underset{4 \times 2}{\overset{A}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \underset{2 \times 3}{\overset{B}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \underset{4 \times 3}{\overset{C}{\begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \\ 7 & -14 & 21 \\ -3 & & \end{bmatrix}}}$$

Dot Product,

$$C_{41} = \overrightarrow{a_4} \cdot \overrightarrow{b_1}$$

$$C_{41} = (-1 \cdot 4) + (1 \cdot 1)$$

$$C_{41} = -3$$

Determine $C = A \cdot B$

$$\overset{A}{\underset{4\times 2}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \overset{B}{\underset{2\times 3}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \overset{C}{\underset{4\times 3}{\begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \\ 7 & -14 & 21 \\ -3 & -5 & \end{bmatrix}}}$$

Dot Product,

$$C_{42} = \overrightarrow{a_4} \cdot \overrightarrow{b_2}$$

$$C_{42} = (-1 \cdot 3) + (1 \cdot -2)$$

$$C_{42} = -5$$

Determine $C = A \cdot B$

$$\overset{A}{\underset{4\times 2}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}}} \cdot \overset{B}{\underset{2\times 3}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}}} = \overset{C}{\underset{4\times 3}{\begin{bmatrix} 11 & 0 & 21 \\ -1 & 13 & -9 \\ 7 & -14 & 21 \\ -3 & -5 & -3 \end{bmatrix}}}$$

Dot Product,

$$C_{43} = \overrightarrow{a_4} \cdot \overrightarrow{b_3}$$

$$C_{43} = (-1 \cdot 6) + (1 \cdot 3)$$

$$C_{43} = -3$$

$$\overset{B}{\begin{bmatrix} 4 & 3 & 6 \\ 1 & -2 & 3 \end{bmatrix}} \cdot \overset{A}{\begin{bmatrix} 2 & 3 \\ 1 & -5 \\ 0 & 7 \\ -1 & 1 \end{bmatrix}} = \overset{C}{\begin{bmatrix} \text{invalid} \end{bmatrix}}$$

2×3 $\quad$ 4×2

columns of A $\neq$ rows of B

Determine

(a) YZ

(b) ZY

$$Y = \begin{bmatrix} 1 & 5 \\ 7 & 2 \end{bmatrix} \qquad Z = \begin{bmatrix} -1 & 3 \\ 9 & 5 \end{bmatrix} \qquad YZ = \begin{bmatrix} 44 & 28 \\ 11 & 31 \end{bmatrix} \qquad ZY = \begin{bmatrix} 20 & 1 \\ 44 & 55 \end{bmatrix}$$

Matrix multiplication is not commutative,

$AB \neq BA$

IDENTITY MATRIX

A **square matrix** in which all the elements of the principal **diagonal** are **ones**, and all other elements are zeros

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Determine

(a) IA

(b) AI

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad A = \begin{bmatrix} 3 & -4 & 7 \\ 8 & 9 & 1 \\ -2 & 5 & 6 \end{bmatrix}$$

$$IA = \begin{bmatrix} 3 & -4 & 7 \\ 8 & 9 & 1 \\ -2 & 5 & 6 \end{bmatrix}$$

$$AI = \begin{bmatrix} 3 & -4 & 7 \\ 8 & 9 & 1 \\ -2 & 5 & 6 \end{bmatrix}$$

Multiplicative Identity

$$I \cdot A = A$$

PROPERTIES OF MATRIX MULTIPLICATION

	Numbers	Matrices
Associative Property	$(ab)c = a(bc)$	$(AB)C = A(BC)$
Commutative Property	$ab = bc$	$AB \neq BC$
Distributive Property	$a(b + c) = ab + ac$	$A(B + C) = AB + AC$
Multiplicative Identity	$1 \cdot a = a$	$I \cdot A = A$

SEATWORK